

Target Library Document

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

1. Target Library Philosophy

Favor mature geospatial and planetary-science tools over custom parsing. The system should be explainable and reproducible, so start with deterministic feature engineering and only add deep learning when data volume and labels justify it.

2. Core Python Libraries

Library	Use	Priority
pds4_tools	Read PDS4 labels and arrays from Chandrayaan-2 products	Must have
rasterio / rioarray	Raster IO, reprojection, clipping, metadata	Must have
xarray / dask	Large raster stacks and chunked computations	High
numpy / pandas	Numerical and tabular processing	Must have
geopandas / shapely / pyproj	Vector geometry, CRS transforms, spatial operations	Must have
scikit-image	Crater/boulder/texture features, morphology, filtering	High
scikit-learn	Baseline scoring, calibration, feature importance	High
networkx	Graph route planning and critical path analysis	Medium
pytorch	Optional CNN/UNet feature extraction if labeled data is available	Stretch
matplotlib / plotly	Diagnostics and scientific plots	High
reportlab	Generate updated PRD, architecture, API/schema, and target-library PDFs	High

3. Planetary and Geospatial Toolchain

Tool	Use	Notes
USGS ISIS	Planetary image calibration, projection, SPICE-aware processing	Best serious choice for planetary raster workflows
NASA Ames Stereo Pipeline	Stereo DEM generation/refinement from imagery	Useful for OHRC/TMC-2 terrain products
GDAL	COG conversion, warping, tiling, raster utilities	Required under most Python geospatial stacks
QGIS	Manual QA, map design, layer inspection	Excellent for demo validation and screenshots
PostGIS	Spatial database and product discovery queries	Recommended backend store
MapLibre GL	Frontend map rendering	Open-source map visualization

Tool	Use	Notes
deck.gl	High-performance raster/vector overlays	Good for heatmaps and path overlays
NASA PGDA / PDS	External terrain and illumination reference datasets	Used for current LOLA validation and future PSR checks

4. Frontend Libraries

- React + TypeScript for the dashboard.
- MapLibre GL for map canvas, layer toggles, bounds, and custom tile sources.
- deck.gl for candidate heatmaps, path overlays, hover picking, and layer blending.
- TanStack Query for API state and caching.
- Zustand or Redux Toolkit for map-layer state if the UI grows.
- Recharts or Plotly.js for evidence charts and score decomposition.
- Guided-tour component or a small state-machine hook for Judge Demo Mode sequencing.

5. Current Demo Feature Dependencies

Feature	Current dependency path	Future library path
Dynamic metrics	Static JSON/JS objects generated from processed layer summaries	FastAPI layer metrics endpoint backed by PostGIS tables
A* traverse	Python-derived route JSON and rendered focus PNG	networkx/custom A* service using COG/Zarr cost grids
LOLA validation	Downloaded PGDA GeoTIFFs cropped with rasterio	Reusable validation job with provenance and tolerance thresholds
Judge Demo Mode	Frontend step state that switches layers and narration	Persisted demo_steps table plus downloadable presenter script
Source ledger	Dashboard and PDF tables from audited source status	Product registry with checksum, source URL, and processing version

External validation now uses NASA PGDA/LRO LOLA south-pole slope, roughness, and terrain-class products cropped to the Chandrayaan-2 TMC-2 AOI. The TMC-derived mean slope is 10.67 deg, the LOLA mean slope is 11.63 deg, and the mean difference is 0.96 deg. Because LOLA is 1000 m/pixel, this is a regional sanity check, not meter-scale hazard validation.

The prototype route from LZ-A to SCI-B is generated with A* over downsampled TMC-2 accessibility and cold-trap proxy rasters. The route has 50 path pixels and a relative cost of 151.74. It is a screening route for judge demonstration and needs geodetic calibration, obstacle inflation, and rover dynamics before mission use.

6. Backend and API Libraries

- FastAPI and Pydantic for typed request/response contracts.
- SQLAlchemy or SQLModel with GeoAlchemy2 for PostGIS tables.
- Celery/RQ or FastAPI BackgroundTasks for heavier preprocessing jobs.
- Uvicorn for local development and Docker Compose deployment.
- ReportLab or WeasyPrint for mission brief PDF generation.

7. Recommended MVP Stack

Part	Pick for Hackathon
Frontend	Vite React + TypeScript + MapLibre GL + deck.gl
Backend	FastAPI + Pydantic + SQLAlchemy
Database	PostgreSQL + PostGIS
Raster storage	Local data directory with Cloud Optimized GeoTIFFs
Processing	Python scripts using pds4_tools, rasterio, geopandas, scikit-image, scikit-learn
Route planning	Grid A* with networkx or a small custom implementation
Validation	NASA PGDA/LRO LOLA crop plus caveated regional comparison
Demo data	Preprocessed AOI bundle downloaded from PRADAN/CH2 Map Browse and NASA PGDA

8. Source-to-Evidence Ledger

Source	Status	Contribution
ISRO PRADAN	Used	Official Chandrayaan-2 DFSAR, TMC-2, and OHRC product downloads.
Chandrayaan-2 Map Browse	Used	AOI/product discovery and south-pole footprint selection.
DFSAR science release	Science anchor	CPR/DOP interpretation target; current prototype uses ratio-based candidate evidence.
TMC-2 south-pole DTM/orthobrowse	Used	Slope, accessibility, hillshade, cold-trap proxy, and route cost layers.
OHRC	Context	High-resolution visual hazard context until exact footprint registration is complete.
NASA PGDA/LRO LOLA	Used for validation	Independent 1000 m/pixel slope, roughness, and class sanity check.
USGS ISIS	Roadmap	Production-grade planetary projection/calibration path documented for later hardening.

9. What Not to Overbuild

- Do not start with a large deep-learning model unless you already have labels and compute.
- Do not store full rasters in PostgreSQL; store files and metadata separately.
- Do not depend on live portal downloads during the final demo.
- Do not claim definitive ice detection; present transparent evidence and uncertainty.

Source Links

ISRO Science Data Archive - Chandrayaan-2 PRADAN: <https://pradan.issdc.gov.in/ch2/>

ISSDC Chandrayaan-2 mission data page: <https://www.issdc.gov.in/chandrayaan2.html>

ISRO DFSAR subsurface-ice release, May 27, 2026: https://www.isro.gov.in/Chandrayaan2_Dual_Frequency_Synthetic_Aperture_Radar.html

Chandrayaan-2 Map Browse: <https://chmapbrowse.issdc.gov.in>

VEDAS note on downloading Chandrayaan-2 OHRC products: https://vedas.sac.gov.in/static/pdf/SIH_2024/SIH1732_CH2_PS.pdf

NASA Planetary Data System Geosciences Node: <https://pds-geosciences.wustl.edu/>

NASA PGDA LRO LOLA south-pole products: <https://pgda.gsfc.nasa.gov/products/90>

USGS ISIS planetary image processing: <https://isis.astrogology.usgs.gov/>